

CMP 405/743: Intro to Networks

Lehman College, CUNY
Spring 2026
Prof. Johnson

Official Course Catalog Description

CMP 405: Intro to Networks: Introduction to network protocols and algorithms. Intensive study of the most important protocols at each layer. Examination of their strengths and weaknesses. Basic algorithms for identifying primary servers, constructing forwarding and broadcasting trees, and determining routing tables. Writing a simple networking service at the I.p. layer or higher. ~~Lab exercises include building and testing small networks.~~ Prereqs: CMP 334 (Computer Organization), CMP 338 (Data Structures and Algorithms I), and CMP 232 (Discrete Structures).

CMP 743: Principles of Communications Networks: Digital and analog communication, system architectures, and connection-oriented and connectionless service. The ~~OSI model~~ [IP protocol stack \("tcp/ip"\)](#) as a conceptual framework, and actual communication models and their protocols. Selected contemporary topics, such as communications security and the World Wide Web. Prereqs: Operation Systems (plus CMP 405's prereqs).

Classroom Rules

1. laptops/phones/etc. during "lecture": "Use only as directed." Keep them put away except when
a) we're all working through a task/demo together on them as part of the lecture or b) I ask you to google/wolframalpha something, etc.
 2. laptops for "labs": please bring your laptop to class every day. Many/most class sessions will have at least a portion of the time devoted to "lab" and/or "us all working through something together".
 3. **no bathrooms** during "lecture": *don't walk out and back into the classroom during lecture.* Being late or having to leave early are both >> leaving & returning.
-

Topics (tentative)

intro: layering, encapsulation, e2e, packet-switching ([reading/listening for next week](#))
http
ip routing/forwarding
tcp, udp
lots of socket programming
dns
dhcp, ...
addressing (IPv4, IPv6, ...)
packet analysis (Wireshark, also devtools)
congestion, fairness, etc.
routing: spanning trees, dijkstra, etc.
ethernet, wifi
network config/analysis (lots of command-line tools)
(http/2? http/3?)

bits of security?

network structure: webgraph, heavy tails, small world, rich get richer, friendship pdox, cascades, viral

search: web crawling, pagerank, etc

net econ: load-balancing & routing games, adwords, net formation games

Recommended texts (*easily obtainable by googling the title and "pdf"*)

- Main networking textbook:
 - [Computer Networking: A Top-Down Approach Featuring the Internet, Kurose and Ross](#)

Alternate text / see also:

- Computer Networks: A Systems Approach (5th edition), Peterson and Davie
- Network programming reference:
 - TCP/IP Illustrated, Volume 1: The Protocols, by Stevens
- Network science:
 - Networks, Crowds, and Markets, Easley and Kleinberg

We're not "following" any one of these, but think of Rurose&Ross as the default text. Regardless of whether there is reading assigned, you'll be able to find coverage of most topics we do in one or more of these texts. Plus there may be some semi-nonformal/nontechnical reading assingments.

Grading (subject to adjustment if, e.g., number of programming assignments changes)

Base score = weighted sum of:

- 40% - ~4 programming assignments:
 - Noncompiling/non-running code == a 0. (The class assumes you can code.)
 - Will be done in Python. (You're *not* expected to already know Python. We'll cover the Python you need.)
 - Some may involve/begin as in-class lab assignments (bring your laptop every day) or quizzes.
- 10% - class participation
- 20% - midterm
- 30% - final

Real score equals:

$$\text{base score} - n \cdot \lfloor \log_2 n \rfloor,$$

where n = # absences. (The mised classes penalty function [values](#), [plotted](#).) Also:

- Being caught on your laptop/phone during lecture = absence.
 - Non-emergency leaving+returning during lecture = absence.
-

People

Instructor: Matthew P. Johnson
~~email~~ whatsapp or discord, tba
office hours: MW 1:45-2:45, G 137d (press buzzer button)

The usual pitch: When you need help, please come to office hours! *Seriously*. It's not bad students who come to office hours; good students to office hours.

Even better is to ask questions / stop me / initiate a discussion *during class*.

Advice

This is a difficult class. Network programming is like normal programming, except much harder. To do well, you'll need to keep up with the class each week, meaning not just studying your notes from class and reading the texts' treatment of that material to whatever degree necessary, but also devoting serious time to the programming assignments. This kind of coding can involve lots of debugging, and it can be very frustrating.

You should expect to spend at least **two hours** studying and doing homework outside of class for **every hour** you spend inside of class. Seriously. Notice this works out to a full course load (12-16) credits corresponding to something in the ballpark of...a full-time jobs: $3 \times 3.67\text{hrs} = 11\text{hrs}$ a week for a 4cr class, meaning 33-44hrs a week total.

If you can't realistically plan to invest that kind of time in the class, you might want to consider delaying the class until a semester when you'll have a lighter load.

I want the course to run smoothly, enjoyably and profitably. Feel free to let me know what you find good, interesting, and fun about the course. Let me know sooner about the opposite.

Collaboration and Academic Integrity

It's a very good idea to form study groups with classmates, and some assignments may permit collaboration (**in which case this will be explicitly stated**).

The internet is very useful for finding other lecture notes, lecture videos, etc. Also, as you may have heard, LLMs exist. The main guidelines are that for any submitted code:

1. You must clearly cite (in a comment in your code) any/all internet/AI/people-who-aren't-me resources that generated more than two contiguous lines of your submitted code.
2. *You must understand the code you submit* (i.e., you should be comfortable explaining what it's doing / how it works if I happen to ask you).